CHE 330 Week 2 Homework

Assigned 9/3/2025, Due 9/9/2025 11:59 PM

1. Define and indicate the formulas and notations for the

   a. Number, Weight, and z-Average Molecular Weights of a polymer

- Number average MW: is the average that reflects the # of molecules present in a sample.

$$\boxed{\bar{M}_n = \frac{\sum_i (N_i M_i)}{\sum_i N_i}}$$

- Weight average MW: is determined by weighing each polymer chain according to its mass. $M_w > M_n$

$$\boxed{\bar{M}_w = \frac{\sum_i (N_i M_i) M_i}{\sum_i w_i} = \frac{\sum_i (N_i M_i^2)}{\sum (N_i M_i)}}$$

Z-average MW:

$$\boxed{\bar{M}_z = \frac{\sum_i (N_i M_i^3)}{\sum (N_i M_i^2)}}$$

↓

is a MW average that strongly emphasizes larger molecules for the polymers.

   b. Polydispersity Index and Degree of Polymerization

- PDI is a measure how broad the molecular distribution of a polymer is.

$$\boxed{PDI : \frac{\bar{M}_w}{\bar{M}_n}}$$

- Degree of polymerization: measures the number of repeating monomer units that are linked together in the polymer chain. =>

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2. During a polymer synthesis experiment, your team members reported the following data for a fractionated polyester made from oxalic acid and 1,6-hexanediol. You have been assigned to compute the Mn, Mw, Mz, PDI and σ of the polyester formed.

	Fractions	1	2	3	4	5	6	7	8	9	sum
m_i	Mass (g)	1.082	0.904	0.801	0.310	0.041	1.043	1.281	1.171	0.241	6.874
M_i	M x10^{-4}	1.042	2.312	2.619	3.283	3.501	3.984	5.017	7.984	9.308	

$$M_n = \frac{\Sigma\, mass(g)}{\Sigma (n_i)} = \frac{\Sigma(mass(g))}{\Sigma\left(\frac{m_i}{M_i} mol\right)} = 27158.593\ g/mol$$

$$n_i = \frac{m_i\, \cancel{g}}{M_i\, \frac{\cancel{g}}{mol}}$$

$$M_w = \frac{\Sigma\, n_i M_i^2\, \frac{g^2}{mol}}{\Sigma\, \underbrace{n_i M_i}_{m(g)}} = \frac{\Sigma\, \underbrace{\frac{m_i}{M_i}}_{n_i} \cdot M_i^2\, \frac{g^2}{mol}}{\Sigma\, m_i\ (g)} = \frac{\Sigma\, m_i \cdot M_i\, \frac{g^2}{mol}}{\Sigma\, m_i\ g} = 41680.482\ g/mol$$

$$PDI = \frac{\bar{M}_w}{\bar{M}_n} = 1.53$$

$$M_z = \frac{\Sigma(n_i \cdot M_i^3)}{\Sigma(n_i M_i^2)} = \frac{\Sigma\left(\frac{m_i}{M_i} \cdot M_i^3\right)}{\Sigma\left(\frac{m_i}{M_i} \cdot M_i^2\right)} = \frac{\Sigma\, m_i \cdot M_i^2\, \frac{g^3}{mol}}{\Sigma\, m_i \cdot M_i\, \frac{g^2}{mol}} = 55728\ g/mol$$

$$\sigma = \bar{M}_n \left(\frac{\bar{M}_w}{\bar{M}_n} - 1\right)^{1/2} = 19859.35\ g/mol$$

excel was used to calculate data:

Mn	27158.593	g/mol	sum of mi	6.874
			sum of mi/Mi	0.000253106
Mw	41680.482	g/mol	sum of miMi	286511.63
			sum of miMi^2	15966731343
PDI	1.534707			
Mz	55728.039	g/mol		
std deviation	19859.357	g/mol		

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3. A polymer mixture P has number-average and weight-average molecular weights of 178,000 and 334,603.93 respectively and was divided into fractions A and B by fractional precipitation.

   a. If A and B have number-average molecular weights of 128,000 and 225,000 respectively, what are the weight fractions of A and B obtained from the initial polymer P?

   b. If A and B have the same polydispersity, what is the polydispersity index?

a) weight fractions : $W_A^* = \frac{W_A}{W_A + W_B \; (W_{total})}$, $W_B^* = \frac{W_B}{W_A + W_B}$

$\bar{M}_{N,A} = 128,000$

$\bar{M}_{N,B} = 225,000$

$\bar{M}_{n,p} = 178,000$

$\bar{M}_{w,p} = 334,603.93$

$\text{Weight}_{tot} = \frac{W_A + W_B}{W_{TOT}}$ W_{TOT}

$1 = W_A + W_B$

$\Rightarrow W_B = 1 - W_A$

$$\bar{M}_{n,p} = \frac{1}{\frac{W_A^*}{M_{n,A}} + \frac{W_B^*}{M_{n,B}}} \Rightarrow \frac{1}{M_{n,p}} = \frac{W_A^*}{M_{n,A}} + \frac{W_B^*}{M_{n,B}}$$

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$$\Rightarrow \frac{1}{178000} = \frac{W_A^*}{128000} + \frac{1 - W_A^*}{225000} \Rightarrow W_A^* = 0.348$$

$$W_B^* = 1 - 0.348 = 0.652$$

b) $\frac{\bar{M}_{w,A}}{\bar{M}_{n,A}} = PDI$, $\frac{\bar{M}_{wB}}{\bar{M}_{n,B}} = PDI$ $\Rightarrow$ $\bar{M}_{w,A} = D \cdot \bar{M}_{N,A}$

$\bar{M}_{wB} = D \cdot \bar{M}_{NB}$

let $PDI = D$

sub

$$\Rightarrow \bar{M}_{wP} = W_A M_{wA} + W_B \cdot M_{wB} \Rightarrow M_{wP} = D\left[W_A M_{NA} + W_B M_{NB}\right]$$

Solve for $D = \frac{M_{wP}}{\left[W_A M_{NA} + W_B M_{NB}\right]} = \frac{334603.93}{0.348(128000) + 0.652(225000)} = 1.75$

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4. What would be Mw, Mn and polydispersity for a sample obtained by mixing 25 g of (A) → polystyrene (Mw = 75,000 g/mol; Mn = 65,670g/mol) with 30 g of another polystyrene (Mw = 88,000 g/mol; Mn = 45,000 g/mol)? → B

$$M_{n\,mix} = \frac{1}{\frac{w_A^*}{M_{wA}} + \frac{w_B^*}{M_{wB}}}$$

$$M_{w\,mix} = w_A^* M_{wA} + w_B^* M_{wB}$$

$$w_A^* = \frac{w_A}{w_A + w_B} = \frac{25g}{25+30} = 0.46$$

$$w_B^* = \frac{w_B}{w_A + B} = \frac{30g}{25+30g} = 0.54$$

≈ 1

$$M_{n\,mix} = \frac{1}{\frac{0.46}{65{,}670\frac{g}{mol}} + \frac{0.54}{45{,}000\frac{g}{mol}}} = 52618.5\ g/mol$$

$$M_{w\,mix} = w_A^* M_{wA} + w_B^* M_{wB} = (0.46 \cdot 75{,}000\ g/mol) + (0.54 \cdot 88000\frac{g}{mol}) = 82020\frac{g}{mol}$$

$$PDI_{mix} = \frac{\bar{M}_{w\,mix}}{\bar{M}_{n\,mix}} = \frac{82020\frac{g}{mol}}{52618.5\frac{g}{mol}} = 1.56$$

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5. Calculate the number-average and weight-average molecular weights of the mixture below, be sure to include the correct formulas for these calculations:

Polymer	Mn (g/mol)	Mw (g/mol)	Weight in Mixture (g)
A	35000	67250	1342
B	48000	58190	3245
C	65548	72549	2458

$$W_A^* = \frac{W_A}{\underbrace{(W_A + W_B + W_C)}_{\Sigma W_{TOT}}} = \frac{1342\,g}{(1342+3245+2458)\,g} = 0.19$$

$$W_B^* = \frac{W_B}{W_{TOT}} = \frac{3245\,g}{(1342+3245+2458)\,g} = 0.46$$

$$W_C^* = \frac{W_C}{W_{TOT}} = \frac{2458\,g}{(1342+3245+2458)\,g} = 0.35$$

$= 1$

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$$\bar{M}_{N\,mix} = \frac{1}{\frac{W_A^*}{M_{NA}} + \frac{W_B^*}{M_{NB}} + \frac{W_C^*}{M_{NC}}} = \frac{1}{\frac{0.19}{35000\,g/mol} + \frac{0.46}{48000\,g/mol} + \frac{0.35}{65548\,g/mol}}$$

$$\bar{M}_{N\,mix} = 49136.40\,g/mol$$

$$M_{W\,mix} = W_A^* \cdot M_{WA} + W_B^* M_{WB} + W_C^* M_{WC} = 0.19\left(67250\,\frac{g}{mol}\right) + 0.46\left(58190\,\frac{g}{mol}\right) + 0.35\left(72549\,\frac{g}{mol}\right)$$

$$\bar{M}_{W\,mix} = 64937.05\,g/mol$$

6. **(Extra Credit)**
For a Mixture of Polymers A and B, prove the following:

$$\overline{M_{w,mix}} = w_A^* M_{w,A} + w_B^* M_{w,B}$$

Start with:

$$w_A^* = \frac{w_A}{w_A + w_B}, \quad \bar{M}_{w,A} = \left(\frac{\Sigma(N_i M_i^2)}{\Sigma(N_i \cdot M_i)}\right)_A, \quad \bar{M}_{w,B} = \left(\frac{\Sigma(N_i \cdot M_i^2)}{\Sigma(N_i \cdot M_i)}\right)_B$$

$$w_B^* = \frac{w_B}{w_A + w_B}$$

$$\bar{M}_{w_{mix}} = \frac{\Sigma(N_i \cdot M_i^2)_A + \Sigma(N_i \cdot M_i^2)_B}{\Sigma(N_i M_i)_A + \Sigma(N_i \cdot M_i)_B}, \quad w_i = \Sigma(N_i M_i) \Rightarrow w_A = \Sigma(N_i M_i)_A, \quad w_B = \Sigma(N_i M_i)_B$$

$$\bar{M}_{w_{mix}} = \frac{\Sigma(N_i M_i^2)_A + \Sigma(N_i M_i^2)_B}{w_A + w_B}, \quad (1) \; w_A = w_A^*(w_A + w_B), \quad w_B = w_B^*(w_A + w_B)$$

$$\bar{M}_{w_{mix}} = \frac{\Sigma(N_i M_i^2)_A + \Sigma(N_i M_i^2)_B}{w_A + w_B}, \quad \bar{M}_w = \frac{\Sigma(N_i M_i^2)}{\Sigma(N_i M_i)} \Rightarrow \Sigma(N_i \cdot M_i^2) = \bar{M}_w \cdot \underbrace{\Sigma(N_i \cdot M_i)}_{w}$$

$$\bar{M}_{w_{mix}} = \frac{\bar{M}_{w_A} \cdot w_A + \bar{M}_{w_B} \cdot w_B}{w_A + w_B} \quad \overset{(1)}{\Longrightarrow}$$

$$\bar{M}_{mix_w} = \frac{\bar{M}_{w_A} \, w_A^*(w_A + w_B) + \bar{M}_{w_B} \, w_B^*(w_A + w_B)}{w_A + w_B}$$

$$\bar{M}_{mix_w} = \frac{\cancel{(w_A + w_B)}\left[w_A^* \bar{M}_{w_A} + w_B^* \bar{M}_{w_B}\right]}{\cancel{(w_A + w_B)}}$$

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$$\Rightarrow \boxed{\overline{M}_{w,mix} = w_A^* \overline{M}_{w,A} + w_B^* \bar{M}_{w,B}}$$